Juliana Londoño Álvarez

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Education and academic positions

Brown University — Providence, USA
STEM Postdoctoral Fellow for Diversity, Inclusion, and Impact — Aug 2024 – present
Division of Applied Mathematics

The Pennsylvania State University — State College, USA
Ph.D. in Mathematics — Aug 2019 – Jun 2024
Thesis: "Attractor-based models for sequences and pattern generation in neural circuits" — Advisor: Carina Curto

Universidad Nacional de Colombia — Medellín, Colombia
B.Sc. in Mathematics — Jan 2013 – Aug 2018
Monograph: "De Rham cohomology and Poincaré duality" — Advisor: Alexander Quintero

Publications

J. Londono Alvarez. Fixed point compositionality via low-rank gluing rules in inhibition-dominated threshold-linear networks. *Preprint available at arXiv.org*.

J. Londono Alvarez, K. Morrison, C. Curto. Attractor-based models for sequences and pattern generation in neural circuits. **Neural Computation**, 2026, Vol. 38, Issue 3, pp. 257–291.

C. Parmelee, **J. Londono Alvarez**, C. Curto, K. Morrison. Sequence generation in inhibition-dominated neural networks. **DSWeb Dynamical Systems Magazine**, 2023.

C. Parmelee, **J. Londono Alvarez**, C. Curto, K. Morrison. Sequential attractors in combinatorial threshold-linear networks. **SIAM Journal on Applied Dynamical Systems**, 2022, Vol. 21, Issue 2, pp. 1597–1630.

Grants and funding

National Science Foundation (NSF) — $270,000
Division of Mathematical Sciences (DMS-2602201) — Aug 2026 – Jul 2029
Sequential and fusion attractors as a mechanism for complex rhythm generation and neural binding in recurrent networks. Single PI.

Invited talks

MathSphere APMA, Brown University, USA	Apr 2026
Carney Lunch Flash, Brown University, USA	Apr 2026
Data Science Seminar, Scuola Internazionale Superiore di Studi Avanzati (SISSA), Italy	Mar 2026
Group for Neural Theory lab meeting, Ecole Normale Supérieure, France	Mar 2026
Math Bio Seminar, University of Iowa, USA	Nov 2025
Boston University Dynamics Seminar, Boston University, USA	Sep 2025
SIAM Conference on Algebraic Geometry, Madison, USA	Jul 2025
SIAM Conference on Dynamical Systems, Denver, USA	May 2025
Special Session of AMS Sectional Meeting, University of Connecticut, USA	Apr 2025
New England Dynamics Seminar, University of Massachusetts Amherst, USA	Nov 2024
Lefschetz Center for Dynamical Systems seminar, Brown University, USA	Nov 2024
2da Semana Institucional de Ciencias Básicas, Fundación Universitaria de Ciencias de la Salud, Colombia	Jul 2024
III Colombian Conference of Applied and Industrial Math, Online, Colombia	Jun 2024
Special Session of AMS Sectional Meeting, Creighton University, USA	Oct 2023
Diversity in Math Bio summer seminar, Society for Mathematical Biology, Online, USA	Jun 2023
Theoretical biology seminar, Pennsylvania State University, USA	Oct 2022
Dynamical Principles of Biological and Artificial Neural Networks Workshop, BIRS, Online, Canada	Jan 2022

Contributed talks

NITMB MathBio Convergence Conference, Chicago, USA — Aug 2025
CCN Junior Theoretical Neuroscientist's Workshop, Flatiron Institute, USA — Jun 2023
CoNNExINS, New York University, USA — Jun 2023

Poster presentations

Computational and Systems Neuroscience (COSYNE), Lisbon, Portugal — Mar 2026
Gordon Research Conference on Robotics, Ventura, USA — Jan 2026
Gordon Research Seminar on Robotics (selected for talk), Ventura, USA — Jan 2026
BRAIN NeuroAI Workshop (selected for Poster Blitz presentation), National Institutes of Health, USA — Nov 2024
Benzon Symposium 67, Copenhagen, Denmark — Sep 2023
Society for Neuroscience, San Diego, USA — Nov 2022
Center for Neural Engineering Retreat, Penn State, USA — Aug 2022
Computational Neuroscience Meeting (CNS), Online, USA — Jul 2021

Awards

AMS Sectional Meeting Graduate Student travel award — Fall 2023
Pritchard Dissertation Fellowship — Fall 2023
COSYNE new attendee travel grant — 2023
Departamental teaching award — Fall 2022

Teaching

Pennsylvania State University, Instructor — Aug 2020–Dec 2022
Courses: MATH 220 – Matrices (3 Fall semesters)

Universidad Nacional de Colombia, Instructor — Jul 2018–Jun 2019
Courses: Vector Geometry (2 semesters)

Universidad Nacional de Colombia, Teaching Assistant — Jan 2017–Jun 2018
Courses: Linear Algebra (1 semester), Vector Geometry (3 semesters)

Professional service

Abstract reviewer, Computational and Systems Neuroscience (COSYNE) Conference — 2025

Professional development

Erdős Institute Spring 2024 Cohort, Online, USA — Spring 2024
Math + Neuroscience Workshop, ICERM, USA — Fall 2023
Mathematical Methods in Computational Neuroscience, Kavli Center, Norway — Summer 2023
Dynamic Brain Workshop, Allen Institute, USA — Summer 2022
Mathematics of Big Data, MSRI, Online, USA — Summer 2021